7.4 Hydroxy Compounds (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	7. Organic Chemistry (A Level Only)
Topic	7.4 Hydroxy Compounds (A Level Only)
Difficulty	Easy

Time allowed: 20

Score: /16

Percentage: /100

Question 1a

Phenol, C_6H_5OH, undergoes many different reactions.

Complete the equations in Fig. 1.1 showing **all** the products of each of these reactions of phenol. If no reaction occurs write **no reaction** in the products box.

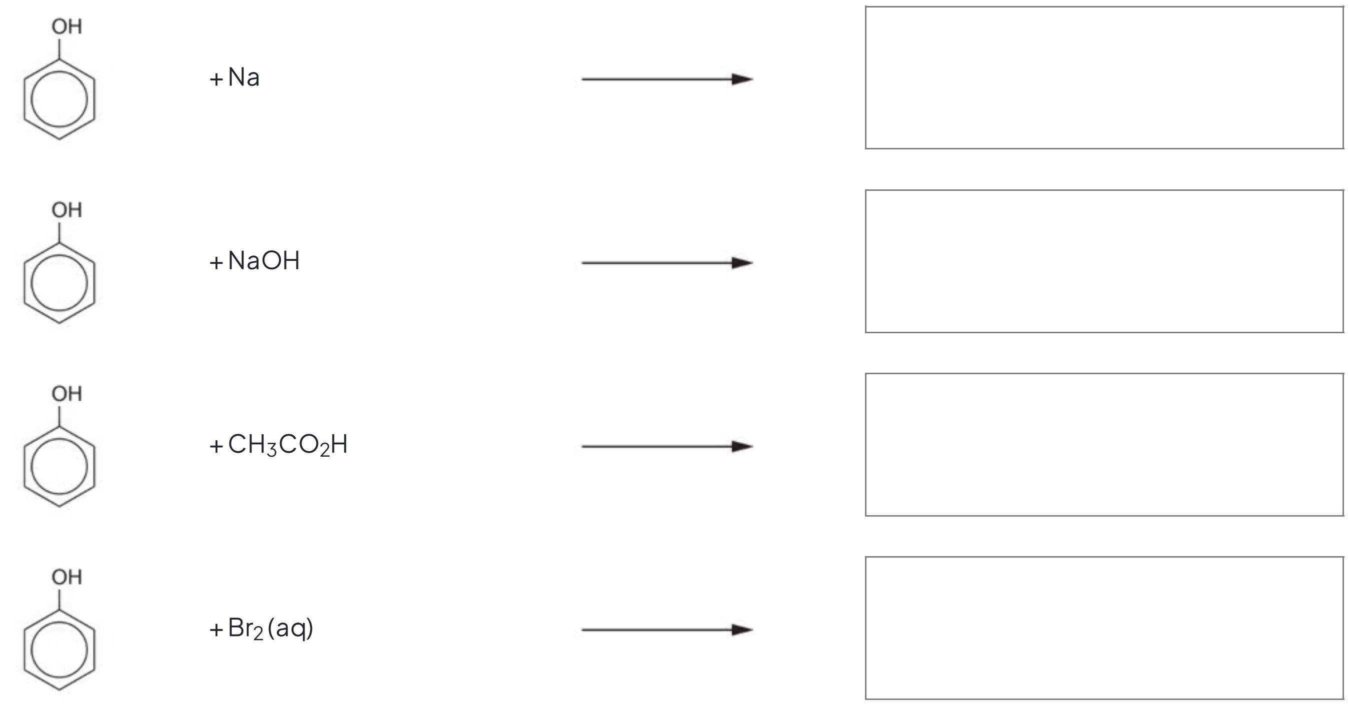


Fig. 1.1

[4 marks]

Question 1b

State the reaction mechanism that occurs when phenol reacts with bromine water.

[1 mark]

Question 1c

Phenol undergoes bromination when reacted with bromine water. Benzene also undergoes bromination but requires different reaction conditions.

State any other necessary conditions for the bromination of phenol and give the necessary reaction conditions for the bromination of benzene.

[2 marks]

Question 2a

Phenol can be prepared by the reaction of phenylamine with nitric(III) acid. This reaction involves three steps.

In the first step, nitric(III) acid, HNO_2, is prepared.

i)

Complete the equation to show the formation of nitric(III) acid.

$$\_\_\_\_\_\_\_\_\_\_\_\_\_ + HCl \rightarrow HNO_2 + \_\_\_\_\_\_\_\_\_\_\_\_\_$$

[1]

ii)
State the necessary condition for this reaction.

[1]

[2 marks]

Question 2b

In the second step in the production of phenol, the diazonium salt shown in Fig. 2.1 is formed. In the final step, the diazonium salt is decomposed by warming with water to produce phenol.

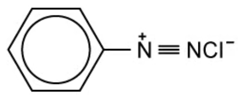


Fig. 2.1

State the formula of any other products.

[2 marks]

Question 2c

Phenol will react with ethanoyl chloride in the presence of sodium hydroxide and heat.

Draw the displayed formula of the organic product and state its name.

[2 marks]

Question 3a

1–naphthol is a phenolic compound. Its structure is shown in Fig. 3.1.

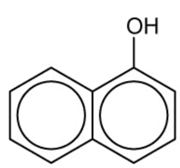


Fig. 3.1

Give the molecular formula of 1–naphthol.

[1 mark]

Question 3b

Like phenol, 1–naphthol is a weak acid.

Write an equation to show how 1–naphthol acts as a weak acid.

[1 mark]

Question 3c

1–naphthol undergoes similar reactions to phenol.

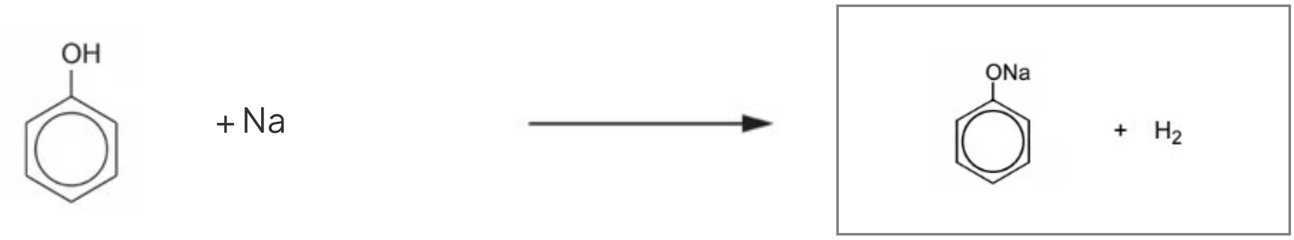
Suggest the structure of the organic compound that is produced when 1–naphthol reacts with bromine water.

[1 mark]

Model Answers: Easy

1a

a) The completed equations showing **all** the products of each of these reactions of phenol are:



- Each correct equation; [1 mark]

[Total: 4 marks]

- Phenol is a weak acid so will act with sodium metal to form the salt sodium phenoxide and hydrogen gas
  - **NOTE**: You do not need to balance this equation for this mark
- Phenol will also react with a base, such as NaOH, to form the salt sodium

phenoxide and water
  - You can show the negative charge on the oxygen and the positive charge on the sodium in sodium phenoxide
- Ethanoic acid is also a weak acid and no reaction will occur between phenol and ethanoic acid
- Phenol reacts with bromine water to form the tri-substituted compound 2,4,6-tribromophenol

1b

b) The reaction mechanism that occurs when phenol reacts with bromine water is:

- Electrophilic substitution; [1 mark]

[Total: 2 marks]

- A lone pair of electrons from the O atom overlaps with the delocalised ring system in the benzene ring, increasing the electron density around the ring
- This makes phenol particularly susceptible to attacks by electrophiles

1c

c) The other necessary conditions for the bromination of phenol and the necessary reaction conditions for the bromination of benzene are:

- Bromination of phenol: Room temperature; [1 mark]
- Bromination of benzene: Pure bromine **AND** aluminium bromide / $AlBr_3$ (catalyst); [1 mark]

[Total: 2 marks]

- Phenol will react more readily with electrophiles as a lone pair of electrons on the O atom overlaps with the pi bonding system of the benzene ring
- This increases the electron density in the ring, making it more susceptible to attacks from electrophiles
- Phenol will react with bromine water at room temperature without a catalyst whereas benzene requires pure bromine (as opposed to a solution) and a catalyst of anhydrous $AlBr_3$

2a

a)

i) The equation to show the formation of nitric(III) acid is:

- $\mathbf{NaNO_2}$ + HCl → HNO_2 + **NaCl**; [1 mark]

ii) The necessary condition for this reaction is:

- Ice / (below) 10 °C; [1 mark]

[Total: 2 marks]

- Nitric(III) acid is unstable so needs to be prepared freshly
- This is usually done in a test tube that is surrounded by ice to keep the temperature below 10 °C

2b

b) When diazonium salt is decomposed by warming with water to produce phenol, the formula of the other products are:

- HCl; [1 mark]
- N_2; [1 mark]

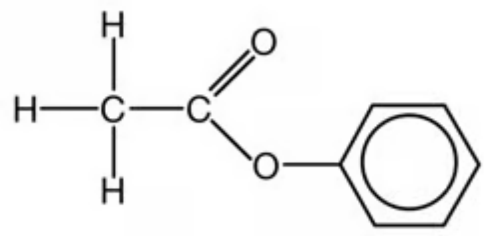
[Total: 2 marks]

- When the diazonium salt is warmed with water, the following reaction occurs:
  - $C_6H_5N_2^+Cl^- + H_2O \rightarrow C_6H_5OH + HCl + N_2$
- You would not be awarded the marks for giving the names of the products as the question specifically asks for the formula

2c

c) The displayed formula of the organic product and its name are:

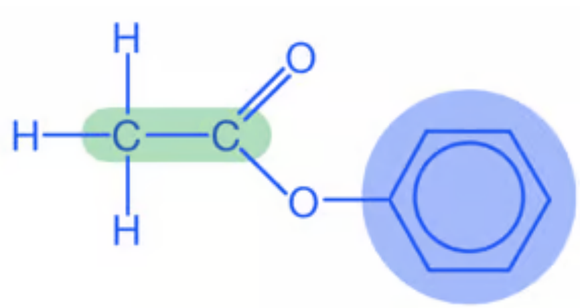
-

 ; [1 mark]

- Phenyl ethanoate; [1 mark]

[Total: 2 marks]

- Acyl chlorides will react with alcohols to produce esters
- The base is required to produce the phenoxide ion $C_6H_5O^-$ which then acts as a nucleophile, attacking the carbonyl carbon in the acyl chloride
- **Remember**: When naming esters, the first part of the name comes from the alcohol used
  - In this case, **phen**ol is used, so the first part of the name is **phen**yl
- The second part of the name comes from the acyl chloride (or carboxylic acid)
  - In this case, **ethan**olyl chlorde is used, so the second part of the name is **ethan**oate



phenyl ethanoate

3a

a) The molecular formula of 1–naphthol is:

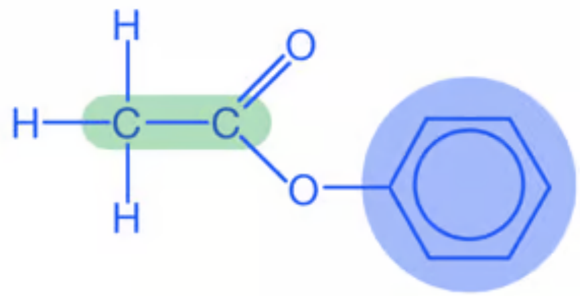
- $C_{10}H_8O$; [1 mark]

[Total: 1 mark]

- It can be useful to sketch 1–naphthol showing the hydrogen atoms
- Careful when counting the carbon atoms, a common mistake is for students to assume there are 12 carbons as there are 2 benzene rings, however, 2 carbon

atoms are shared between the 2 rings giving a total of 10 carbons

ethanoate



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3a

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- $C_{10}H_8O$; [1 mark]

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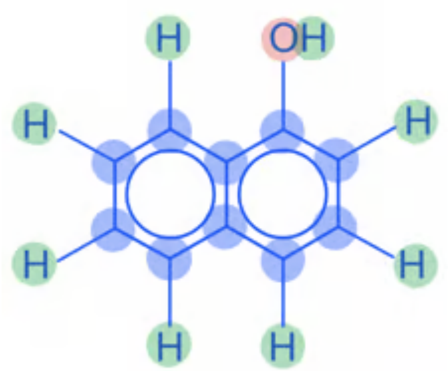
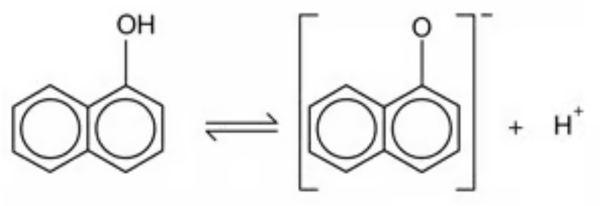


10 carbons atoms, 8 hydrogen atoms, 1 oxygen atom

3b

b) An equation to show how 1–naphthol acts as a weak acid is:



OR

$C_{10}H_7OH \rightleftharpoons C_{10}H_7O^- + H^+$

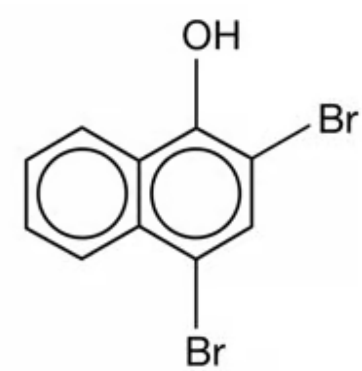
- Correct formulae **AND** use of reversible symbol; [1 mark]

[Total: 1 mark]

- You would not be penalised if you did not use square brackets
- The square brackets show that the charge is spread across the whole ion as one of the lone pairs of electrons on the oxygen atom overlaps with the delocalised π system of the ring
- You could write the equation with water, i.e. $C_{10}H_7OH + H_2O \rightleftharpoons C_{10}H_7O^- + H_3O^+$

3c

c) The structure of the organic compound that is produced when 1–naphthol reacts with bromine water is:

- 

; [1 mark]

[Total: 1 mark]

- When phenol reacts with bromine water, 2,4,6–tribromophenol is produced
  - The -OH group is activating so directs incoming electrophiles to the 2,4 and 6 positions
- A similar reaction occurs with 1–naphthol, except position 6 is not possible as there is no hydrogen on this carbon so it cannot be substituted with a bromine atom